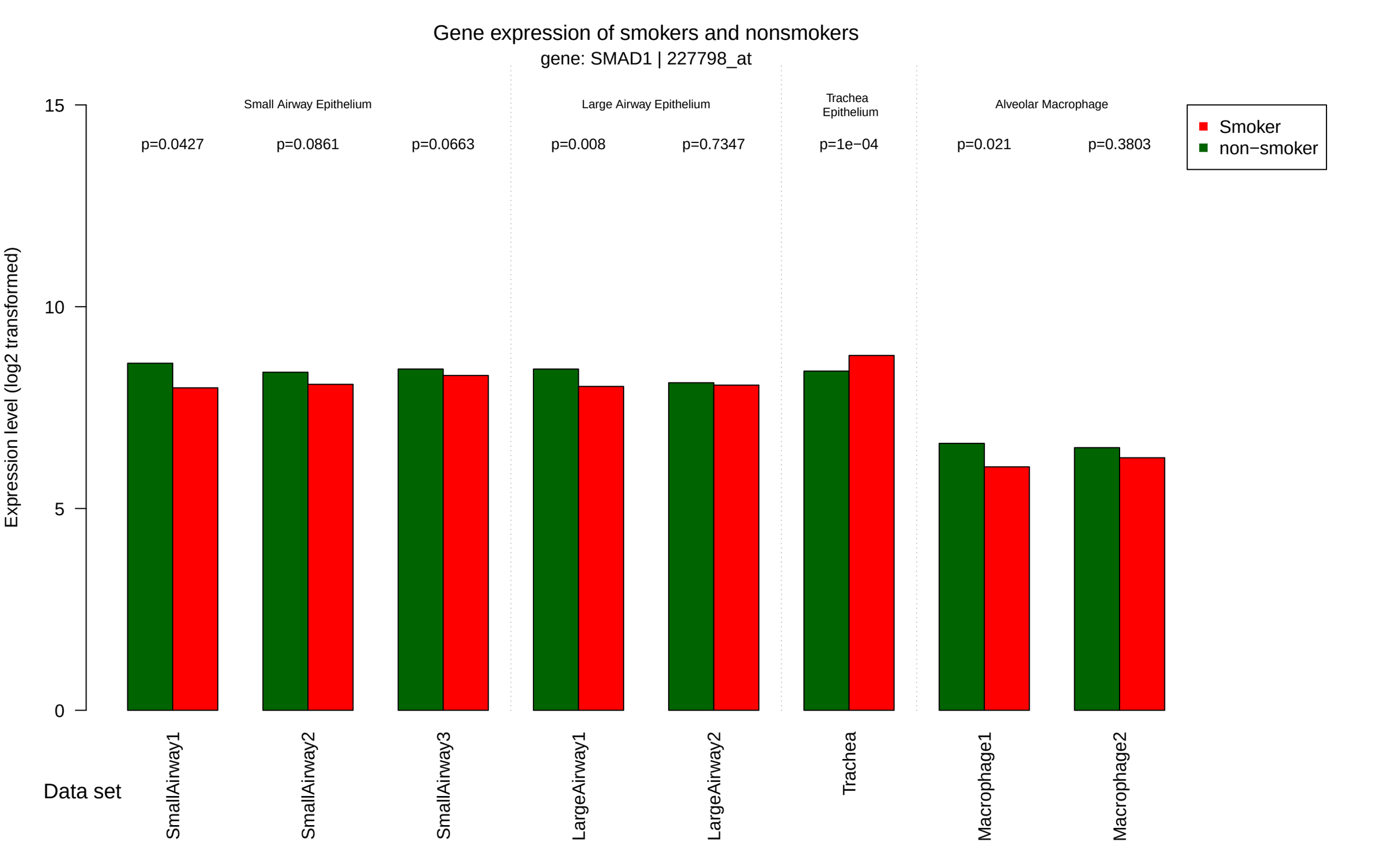
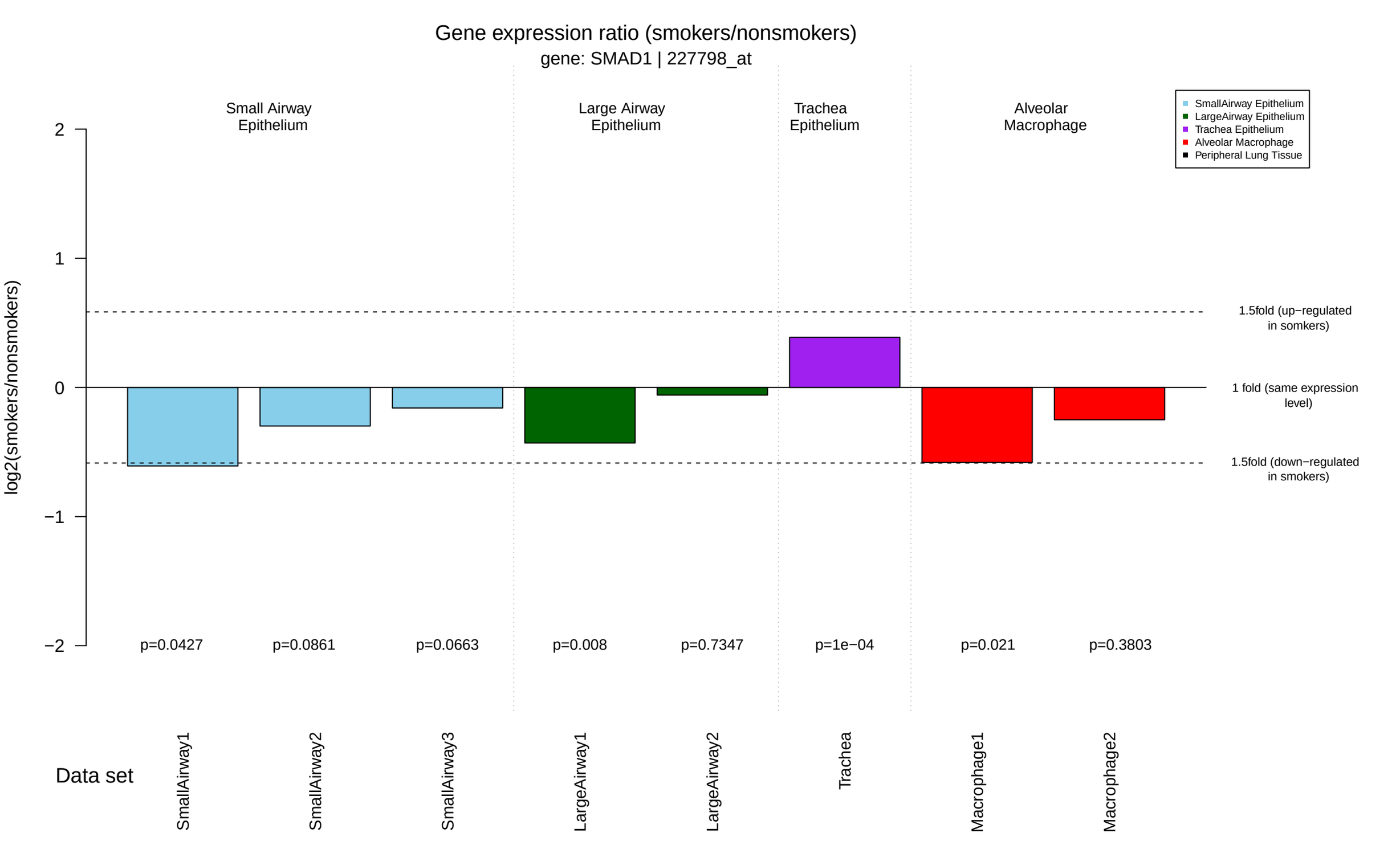
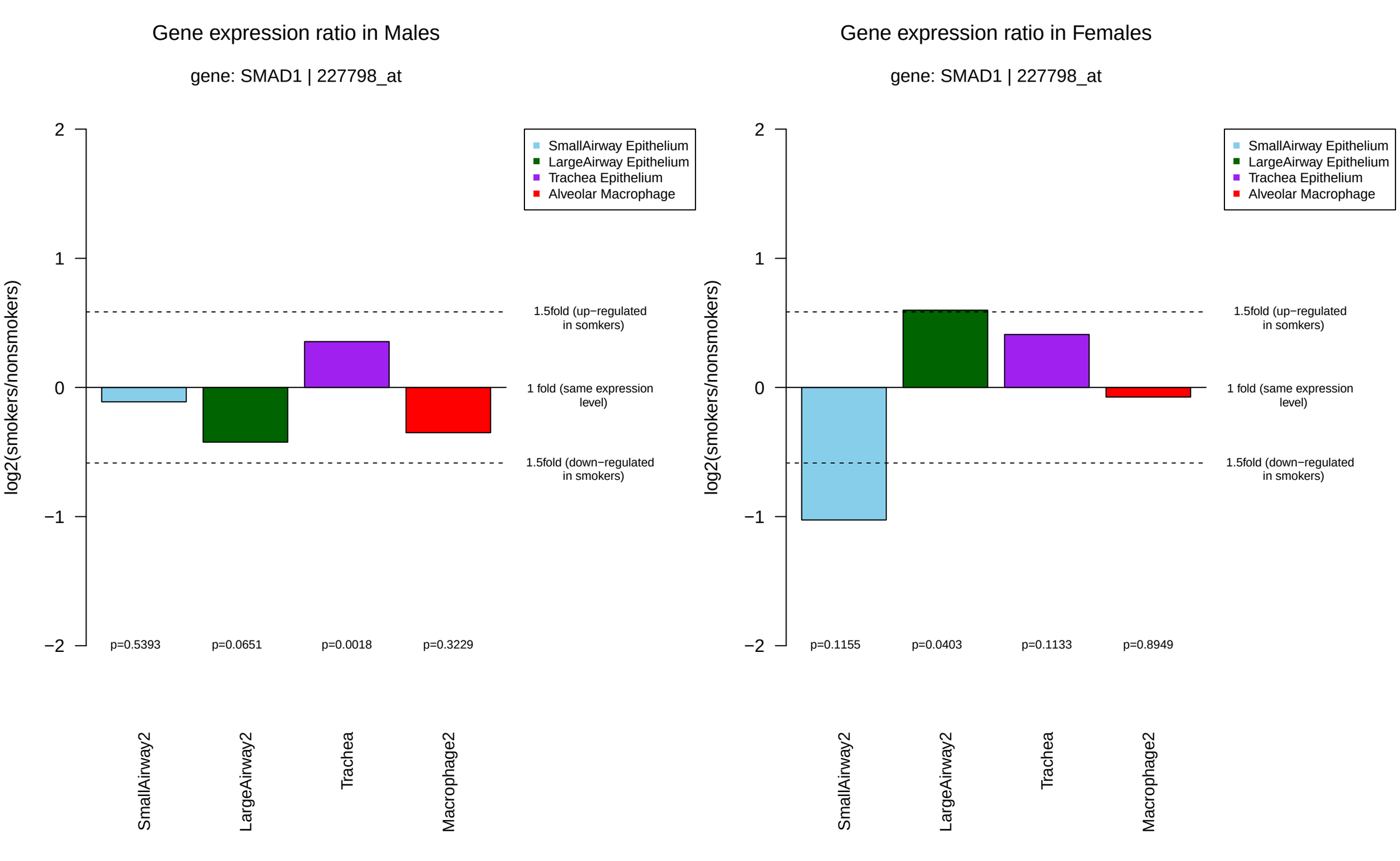
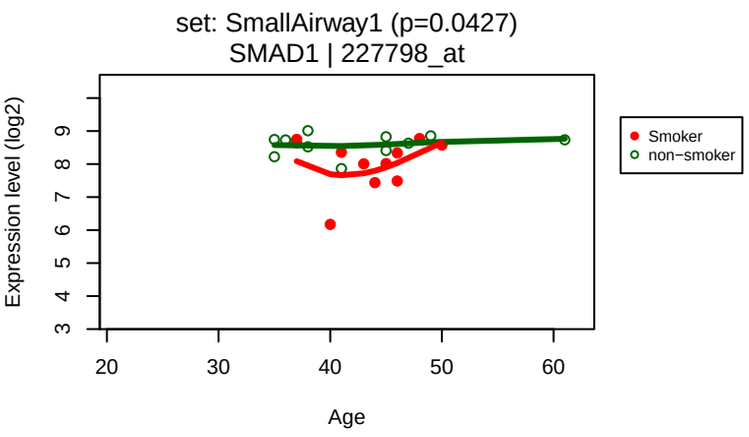
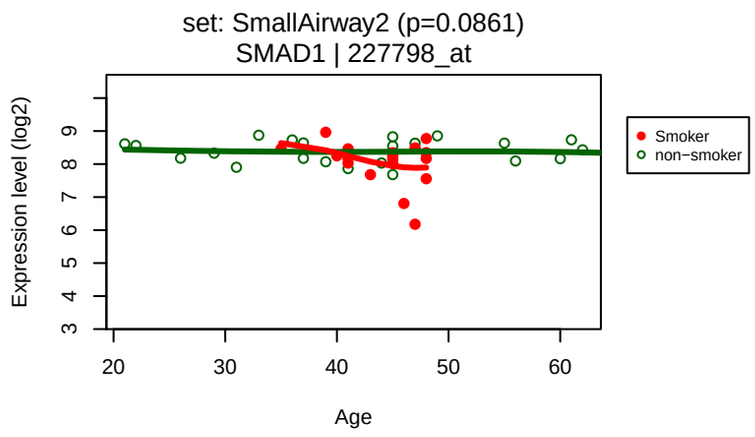
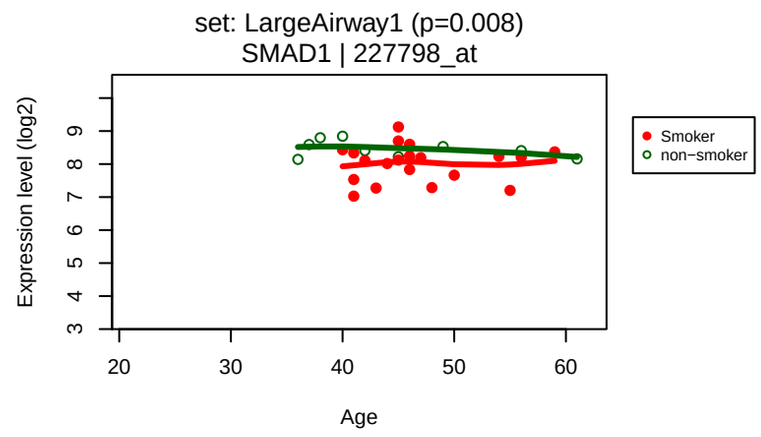

Gene expression of smokers and nonsmokers
gene: SMAD1 | 227798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0427
p=0.0861
p=0.0663
p=0.008
p=0.7347
p=1e−04
p=0.021
p=0.3803
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

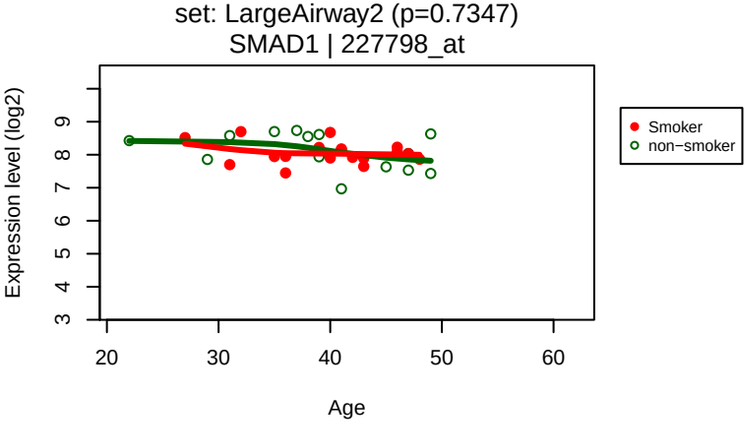

Gene expression ratio (smokers/nonsmokers)
gene: SMAD1 | 227798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0427
p=0.0861
p=0.0663
p=0.008
p=0.7347
p=1e−04
p=0.021
p=0.3803
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

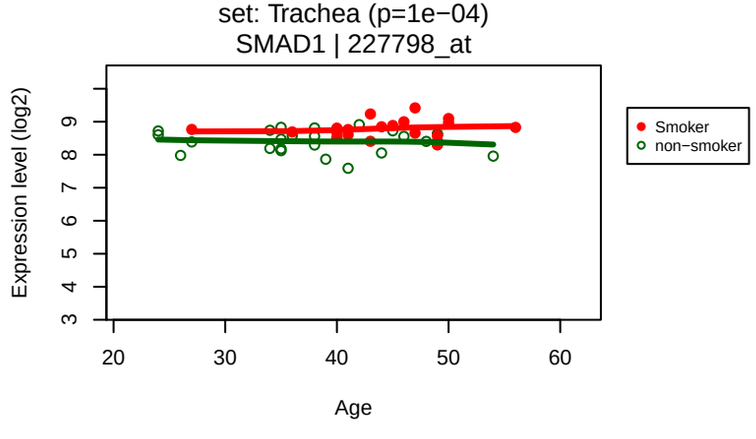

Gene expression ratio in Males
gene: SMAD1 | 227798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5393
p=0.0651
p=0.0018
p=0.3229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD1 | 227798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1155
p=0.0403
p=0.1133
p=0.8949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

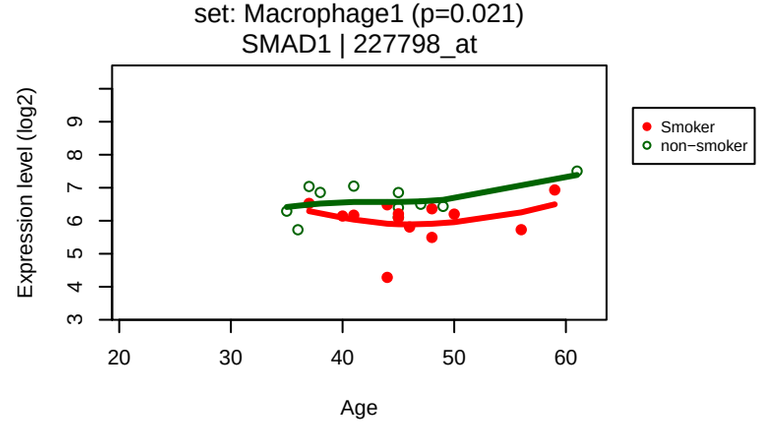

set: SmallAirway1 (p=0.0427)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker

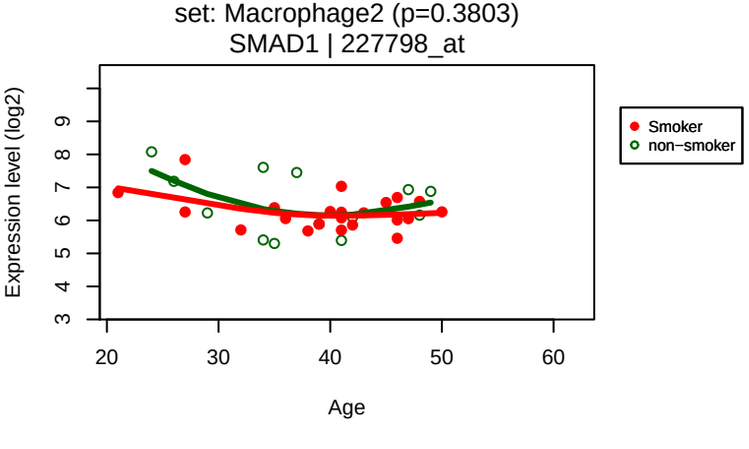

set: SmallAirway2 (p=0.0861)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker

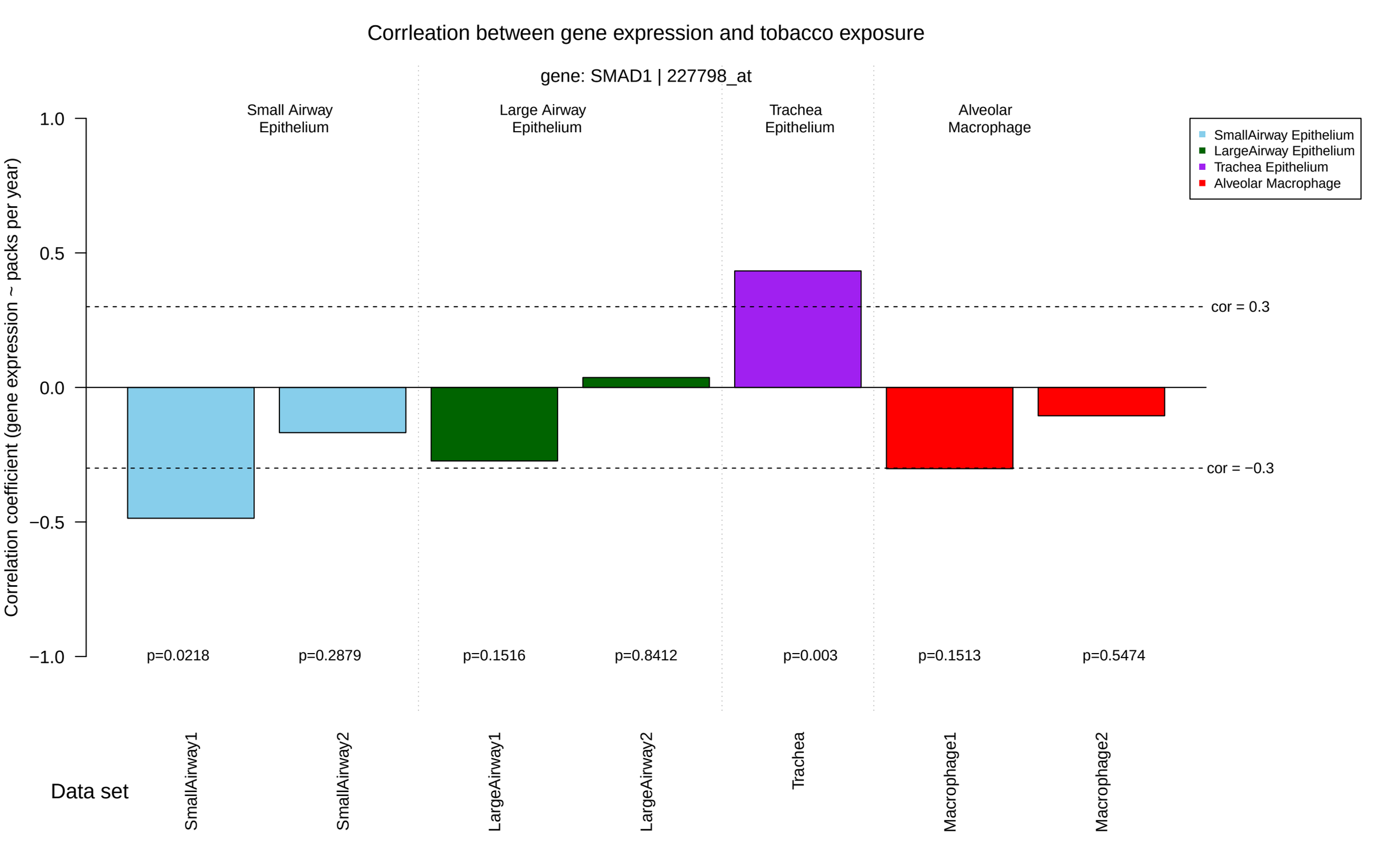

set: LargeAirway1 (p=0.008)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7347)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.021)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3803)
SMAD1 | 227798_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD1 | 227798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0218
p=0.2879
p=0.1516
p=0.8412
p=0.003
p=0.1513
p=0.5474
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2